

REVIEW PAPER

Comparative Analysis of Osimertinib and BDTX-1535 Binding Affinities Against Wild-Type and Mutated Epidermal Growth Factor Receptor (EGFR kinase domains)

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Abstract

Drug resistance is a major problem for non-small cell lung cancer (NSCLC) patients taking Osimertinib, a common third-generation drug. Each new drug generation is developed to overcome mutations that earlier drugs could not handle—often because of changes in the ATP hinge region or shifts between covalent and non-covalent binding. In this computational study, we used molecular docking to compare how Osimertinib and a fourth-generation drug, BDTX-1535, bind to EGFR. We looked at both the wild-type receptor and four different mutant versions. To choose which mutations to study, we used genomic databases like cBioPortal to find the most common ones found in human populations. We then retrieved the corresponding 3D protein structures from the Protein Data Bank (PDB) and drug structures from PubChem. For docking, we ran simulations with CB-Dock, which itself uses AutoDock Vina. These tools model how the ligands and receptors interact and give us binding affinity scores in Gibbs free energy (ΔG , kcal/mol). By comparing the predicted binding poses across the wild-type and mutant receptors, we evaluated how well BDTX-1535 might work against resistant forms. Overall, our results remained null due to several factors and inappropriate comparison. We assumed AutoDock Vina and CB-Dock were two separate platforms with differences lying within the covalent and non-covalent binding between the ligand and protein, however that was not true. Usage of CB-Dock's automated binding site detection was also used while in AutoDock Vina we inputted manual coordinates following 3 amino acids to locate the binding site. Thus comparison may have been imprecise, leading to a null trend. We recommend pair free-energy calculations along with running simulations for higher accuracy.